

Cross-Browser Proxy Unmasking

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Content

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It's really time to sleep in my timezone, but I just couldn't resist when I read latest RSnake's post about [Deanonymizing Tor and Detecting Proxies](#).

The basic concept, not terribly new by the way, is that browser proxy settings cannot be enforced on browser plugins, which happily ignore them in some circumstances, e.g. when establishing a direct TCP socket connection. So if you're using a proxy to hide your internet address (like [Tor](#) users do), embeddable objects like Java applets can betray you, revealing your real identity to advertisers spying on your habits or, worse, to the police of a repressive state.

This caveat has been preached even [on the Tor download page](#) itself, but nothing better than some scary demos to convert the non believers.

RSnake's interesting [proof of concept](#) exploits JavaScript + LiveConnect , and it apparently works on Gecko-based browser with Java™ installed only. I didn't manage to make it work on Opera, even though it does support LiveConnect.

So I decided to defer bedtime a bit and I put together [my own quick deanonymizing proof of concept](#), which relies on the almost ubiquitous Macromedia® Flash® and works in any web browser, like Internet Explorer, supporting the Flash player (no need for JavaScript, either). The `XMLSocket` ActionScript object is used to bypass browser's proxy setting and connect to a tiny server written in Perl, listening on port 9999 and echoing client's IP.

Here's the ActionScript code:

```
[viewcode]src="http://evil.hackademix.net/proxy_bypass/pbp.as" geshi=javascript link=yes scroll=yes[/viewcode] And here's the Perl server:
```

```
[viewcode]src="http://evil.hackademix.net/proxy_bypass/pbp.pl" geshi=perl link=yes scroll=yes[/viewcode] Today's lesson is: if you want to stay anonymous, you'd better turn off Java, Flash and any other plugin!
```

Update OCT-27

I've just learned that [some months ago](#) a guy called **yawnmoth** demonstrated an [Unmasking Java Applet](#). Just like [my Flash-based one](#), this works also in browsers, like IE, not supporting LiveConnect. The [lesson above](#) obviously applies, even stronger.

Demos

- [Cross-Browser Proxy Unmasking Demo in Flash](#)
- [Cross-Browser Proxy Unmasking Demo in Java](#)
- [Gecko-only Proxy Unmasking Demo in JavaScript+Java \(LiveConnect\)](#)

[image: image]

By ma1

Hacker, atheist, humanist, dad, mozillian, security breaker and builder, creator of NoScript, casting spells at the Tor Browser. He/him.

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